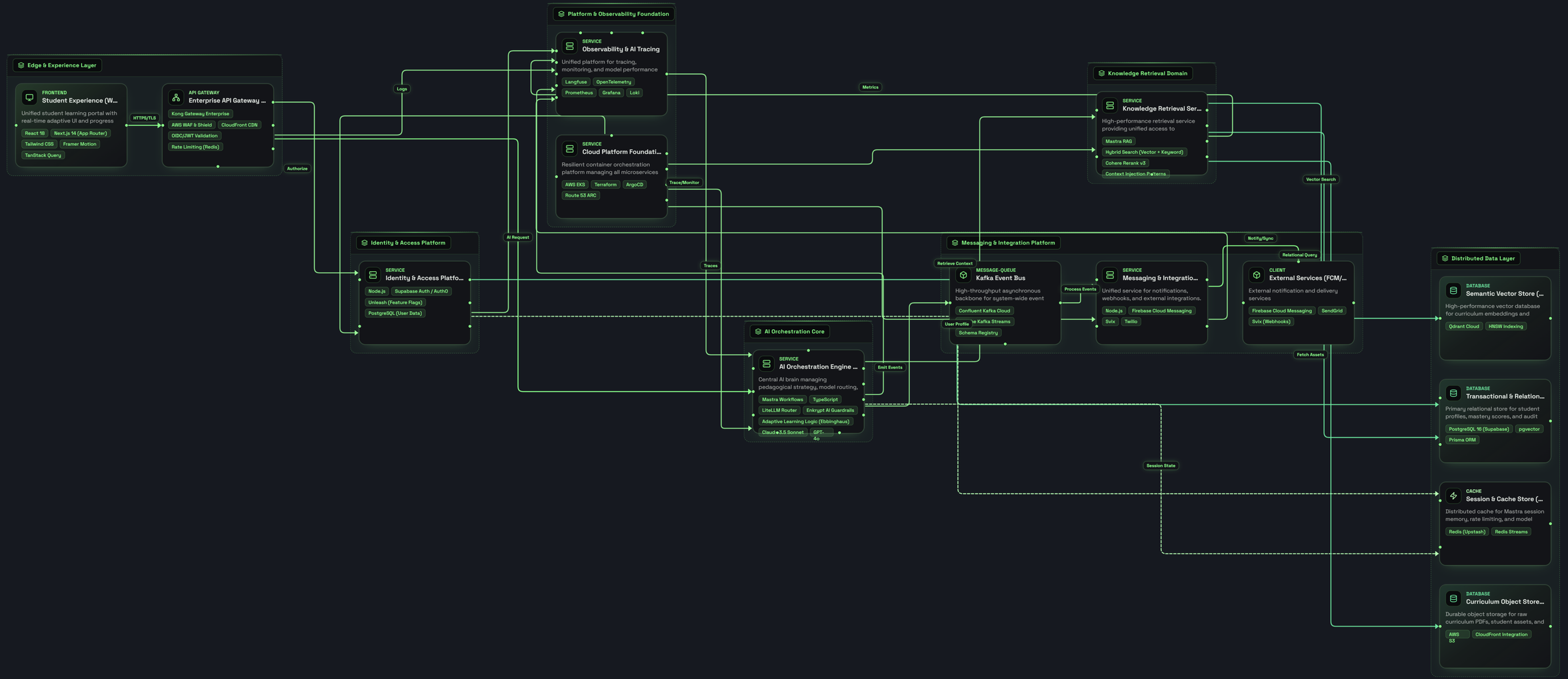
Edge & Experience Layer
FRONTEND
Student Experience (W...
Unified student learning portal with real-time adaptive UI and progress
React 18
Next.js 14 (App Router)
Tailwind CSS
Framer Motion
TanStack Query
HTTPS/TLS
API GATEWAY
Enterprise API Gateway ...
Kong Gateway Enterprise
AWS WAF & Shield
CloudFront CDN
OIDC/JWT Validation
Rate Limiting (Redis)
Authorize
Platform & Observability Foundation
SERVICE
Observability & AI Tracing
Unified platform for tracing, monitoring, and model performance
Langfuse
OpenTelemetry
Prometheus
Grafana
Loki
SERVICE
Cloud Platform Foundati...
Resilient container orchestration platform managing all microservices
AWS EKS
Terraform
ArgoCD
Route 53 ARC
Logs
Metrics
Trace/Monitor
Traces
Knowledge Retrieval Domain
SERVICE
Knowledge Retrieval Ser...
High-performance retrieval service providing unified access to
Mastra RAG
Hybrid Search (Vector + Keyword)
Cohere Rerank v3
Context Injection Patterns
Vector Search
Identity & Access Platform
SERVICE
Identity & Access Platfo...
Node.js
Supabase Auth / Auth0
Unleash (Feature Flags)
PostgreSQL (User Data)
AI Request
User Profile
Messaging & Integration Platform
MESSAGE-QUEUE
Kafka Event Bus
High-throughput asynchronous backbone for system-wide event
Confluent Kafka Cloud
Kafka Streams
Schema Registry
Retrieve Context
Process Events
SERVICE
Messaging & Integratio...
Unified service for notifications, webhooks, and external integrations.
Node.js
Firebase Cloud Messaging
Svix
Twilio
Notify/Sync
Relational Query
CLIENT
External Services (FCM/...
External notification and delivery services
Firebase Cloud Messaging
SendGrid
Svix (Webhooks)
Fetch Assets
AI Orchestration Core
SERVICE
AI Orchestration Engine ...
Central AI brain managing pedagogical strategy, model routing,
Mastra Workflows
TypeScript
LiteLLM Router
Enkrypt AI Guardrails
Adaptive Learning Logic (Ebbinghaus)
Claude 3.5 Sonnet
GPT-4o
Emit Events
Session State
Distributed Data Layer
DATABASE
Semantic Vector Store (...
High-performance vector database for curriculum embeddings and
Qdrant Cloud
HNSW Indexing
DATABASE
Transactional & Relation...
Primary relational store for student profiles, mastery scores, and audit
PostgreSQL 16 (Supabase)
pgvector
Prisma ORM
CACHE
Session & Cache Store (...
Distributed cache for Mastra session memory, rate limiting, and model
Redis (Upstash)
Redis Streams
DATABASE
Curriculum Object Store...
Durable object storage for raw curriculum PDFs, student assets, and
AWS S3
CloudFront Integration